Newton's Rings

Physics Lab 4.18

1. Objective

Measure the radius of curvature of a plano-convex lens from the diameters of its Newton's rings, and quantify how much of the result comes from the microscope rather than from the optics.

2. Theory

The air film between the lens and the flat plate has thickness $e(r)$; a dark fringe appears where $2e + \lambda/2 = k\lambda$. For the k-th dark ring of diameter d_k the surface obeys

$$R = \frac{d_m^2 - d_n^2}{4\,(m-n)\,\lambda}$$

with $\lambda = 589.3\ \text{nm}$ for sodium light. Taking the difference of two rings cancels the unknown film thickness at the contact point, so only the *slope* of d_k^2 against k matters:

$$\frac{\mathrm{d}(d^2)}{\mathrm{d}k} = 4R\lambda$$

3. Measurements

Ring k	Left / mm	Right / mm	d_k / mm	d_k^2 / mm²
10	15.732	12.318	3.414	11.655
11	15.802	12.243	3.559	12.667
12	15.874	12.168	3.706	13.734
13	15.946	12.092	3.854	14.853
14	16.020	12.014	4.006	16.048
15	16.096	11.937	4.159	17.297
16	16.174	11.858	4.316	18.628
17	16.255	11.778	4.477	20.043
18	16.338	11.696	4.642	21.548
19	16.423	11.612	4.811	23.146

4. Fit

A linear least-squares fit of d_k^2 against k is more honest than differencing pairs, because it reports the residual:

```
import numpy as np

k = np.arange(10, 20)
d2 = np.array([11.655, 12.667, 13.734, 14.853, 16.048,
               17.297, 18.628, 20.043, 21.548, 23.146])
slope, intercept = np.polyfit(k, d2, 1)
R = slope / (4 * 589.3e-6)          # mm
resid = d2 - (slope * k + intercept)
print(f"slope = {slope:.4f} mm^2/ring, R = {R:.1f} mm, max resid = {abs(resid).max():.3f}")
```

The fit gives a slope of $1.276\ \text{mm}^2/\text{ring}$, hence $R \approx 541\ \text{mm}$ with a maximum residual of $0.19\ \text{mm}^2$ — a 3% scatter dominated by reading the drum vernier to the nearest 0.005 mm.

5. Discussion

1. **Differencing versus fitting.** The pair-difference method and the least-squares fit agree to within 1%; the fit additionally exposes outliers, which is why it is the better choice when the report has to defend its numbers.
2. **Where the error lives.** A 0.01 mm error in one reading shifts d^2 by about 0.03 mm², roughly 2% of the slope. Rings below $k = 10$ are too wide to locate precisely; rings above $k = 25$ sit too close together.
3. **Systematic term.** Pressure on the lens flattens the contact point and shifts every d^2 by the same offset — an additive constant, which is exactly what the intercept of the fit absorbs.